

Évaluer les performances d'un système asservi à partir d'un modèle de connaissance

Travaux Dirigés

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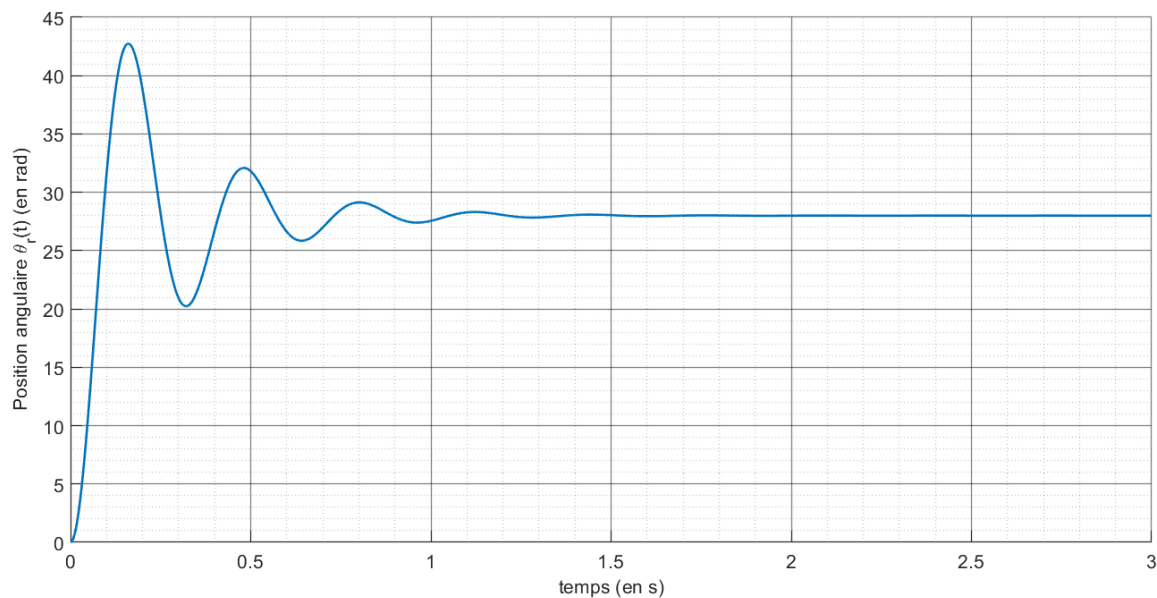
TD 4 : Exercices supplémentaires**Exercice 4.1 : Analyse de réponses temporelles**

Toutes les réponses temporelles ci-dessous sont issues d'un système soumis à un échelon unitaire montant ($0 \rightarrow 1$) ou descendant ($1 \rightarrow 0$).

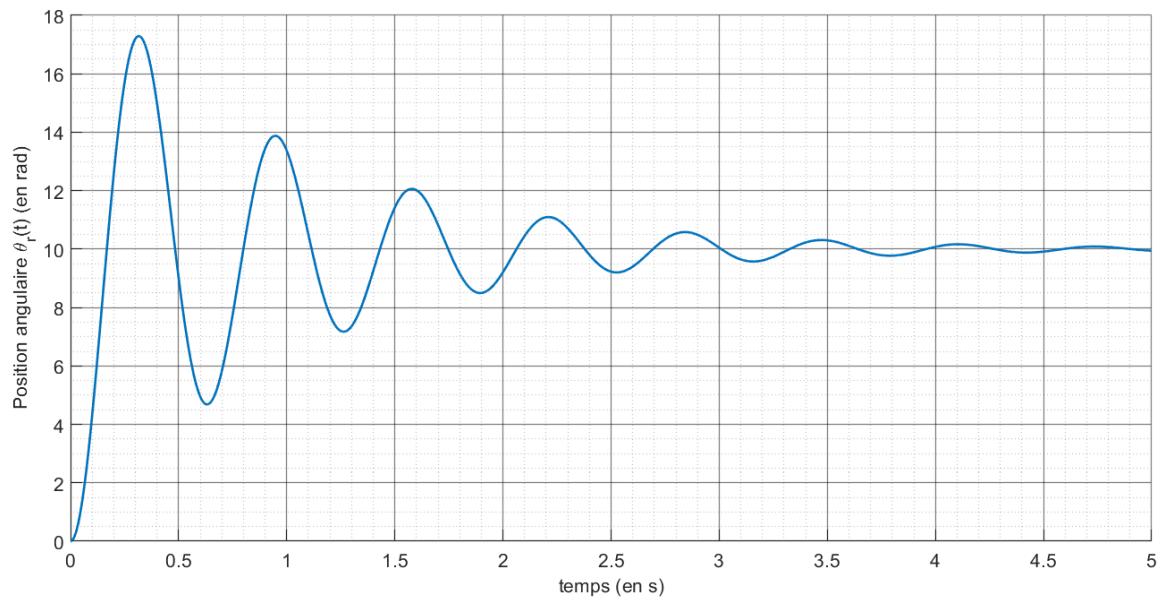
Pour chaque réponse temporelle, déterminer :

- 1- Si la réponse temporelle est la conséquence d'un échelon montant ou descendant ;
- 2- Le gain statique noté K ;
- 3- Avec les abaques, le coefficient d'amortissement z ;
- 4- La pulsation propre noté ω_0 ;
- 5- Avec les abaques, le temps de réponse à 5% noté $tr_{5\%}$.

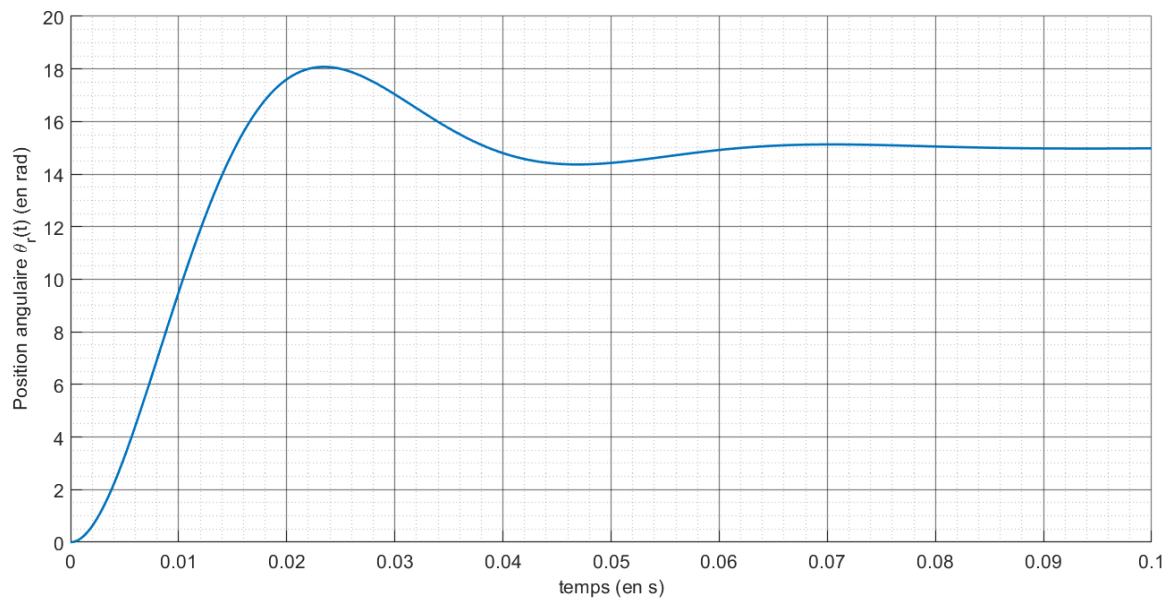
Q1 : Système 1 :



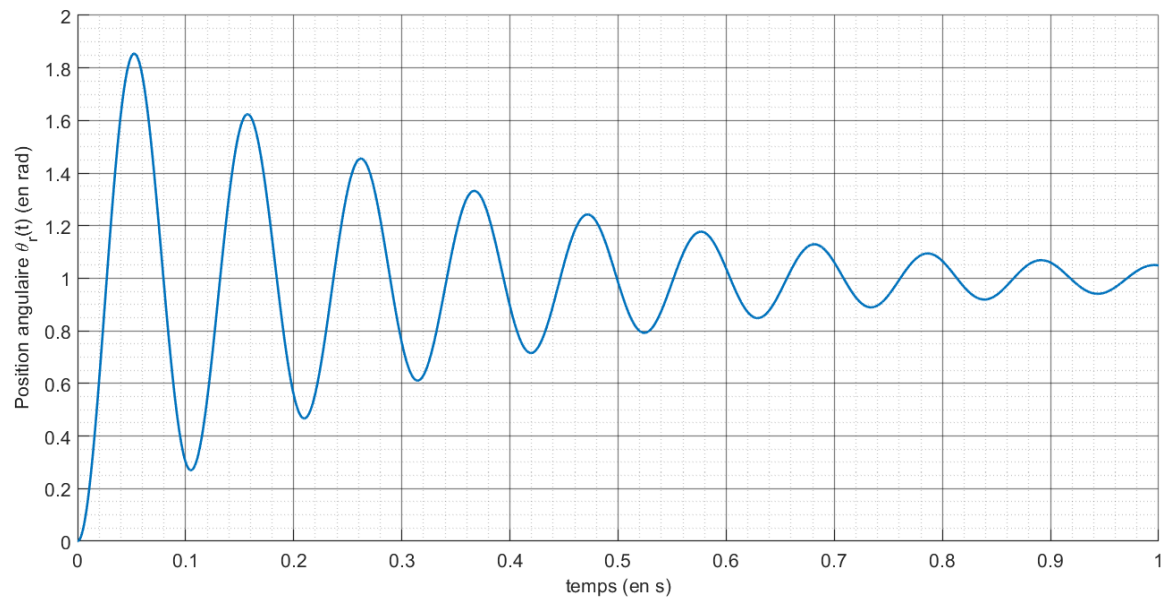
Q2 : Système 2 :



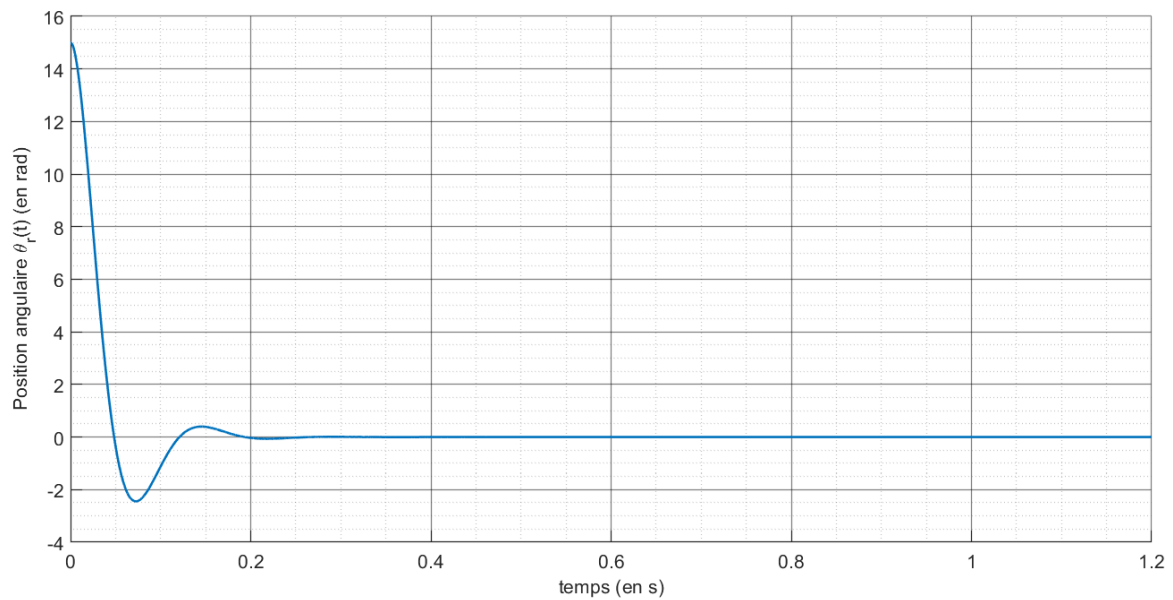
Q3 : Système 3 :



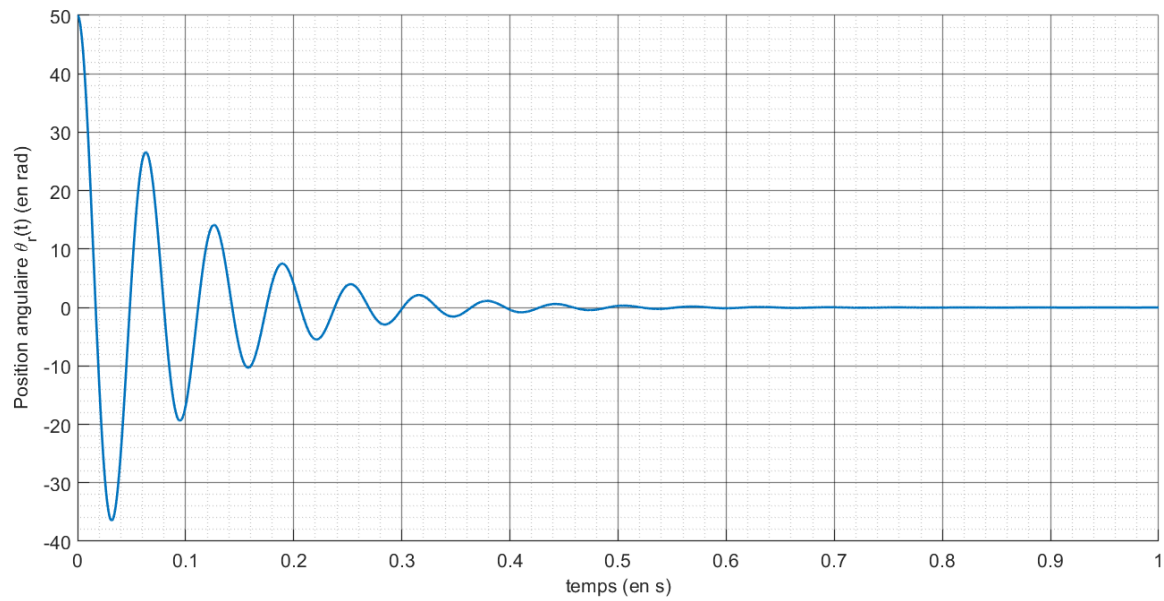
Q4 : Système 4 :

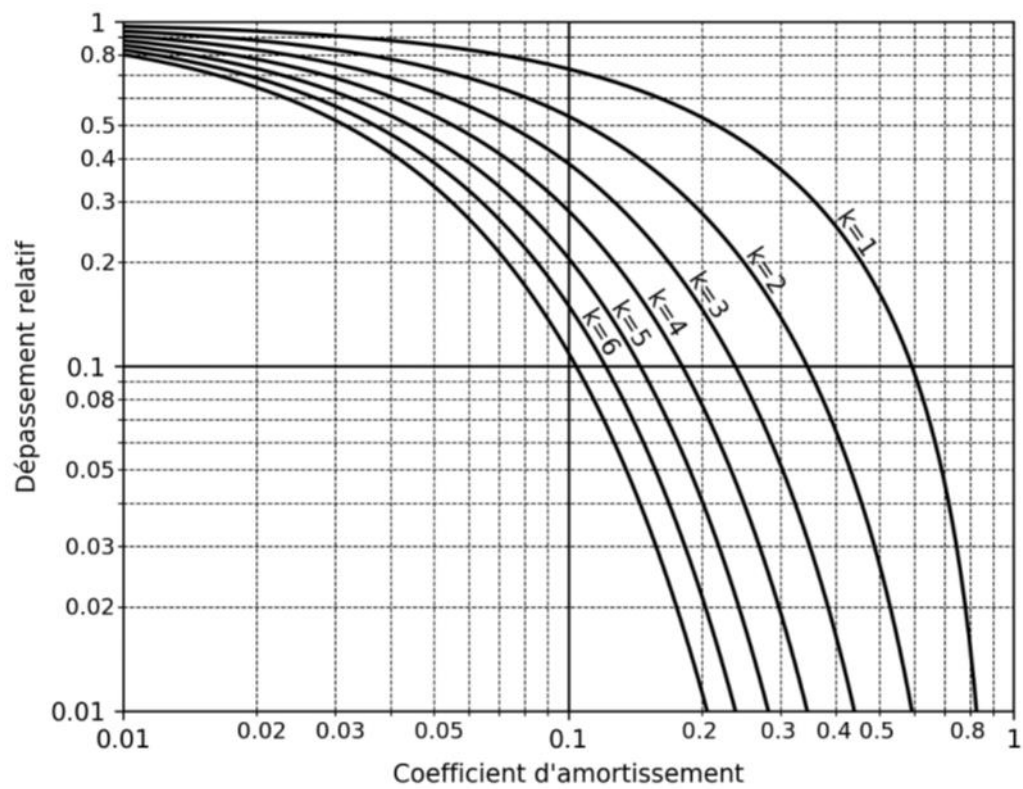


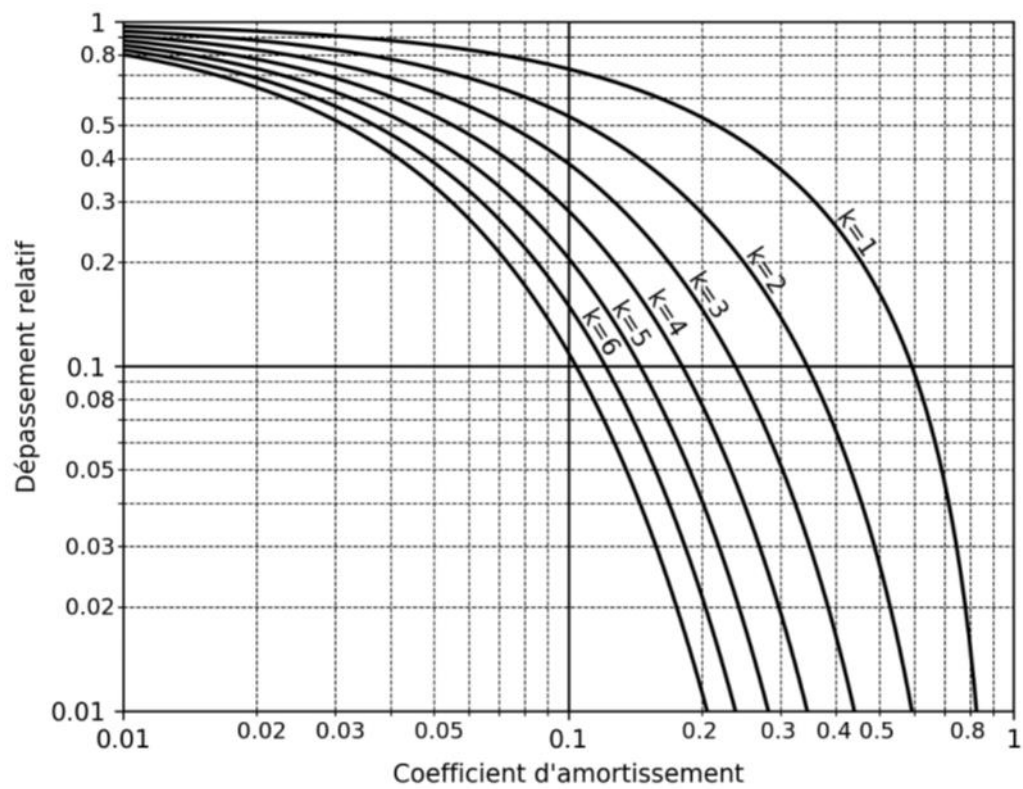
Q5 : Système 5 :

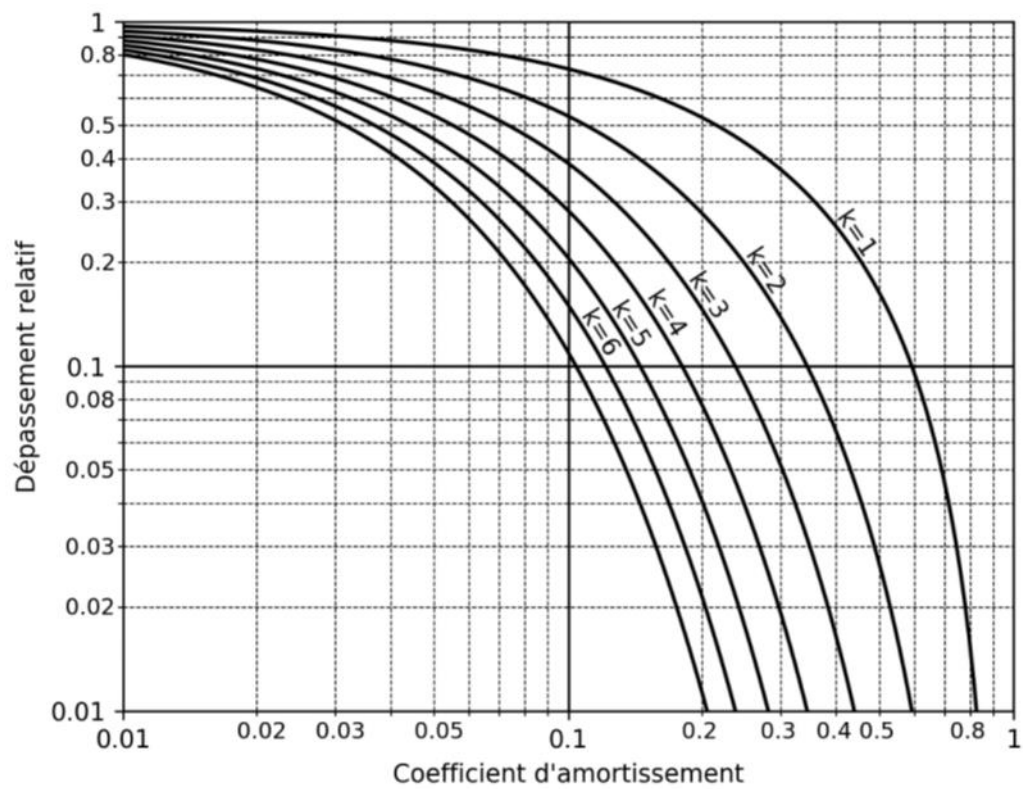


Q6 : Système 6 :





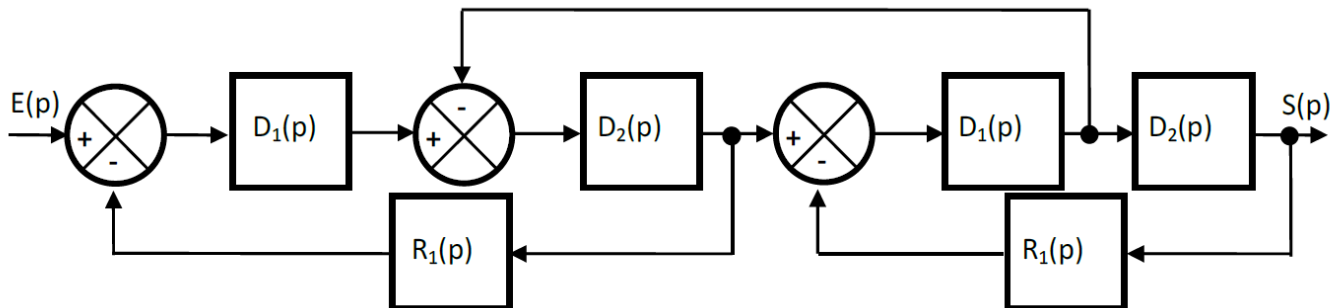




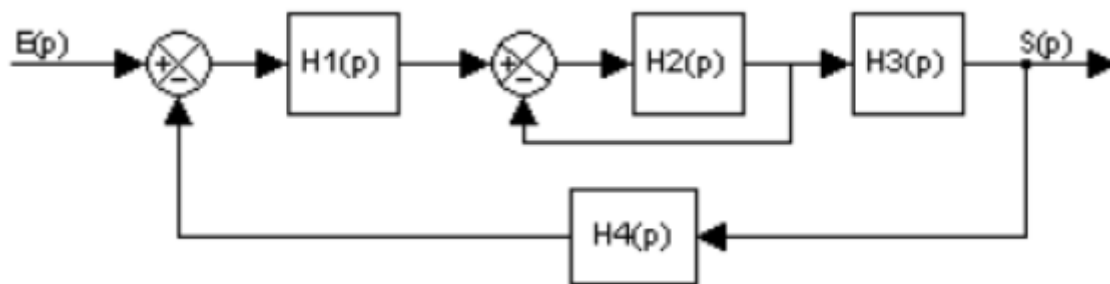
Exercice 4.2 : Fonction de transfert – Schéma bloc

Déterminer la relation entrée / sortie des schémas bloc suivant :

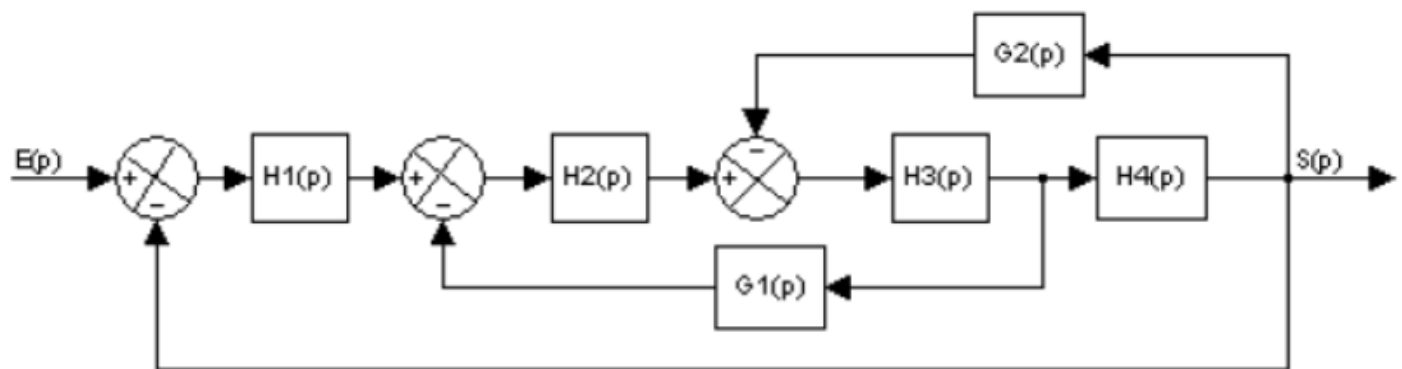
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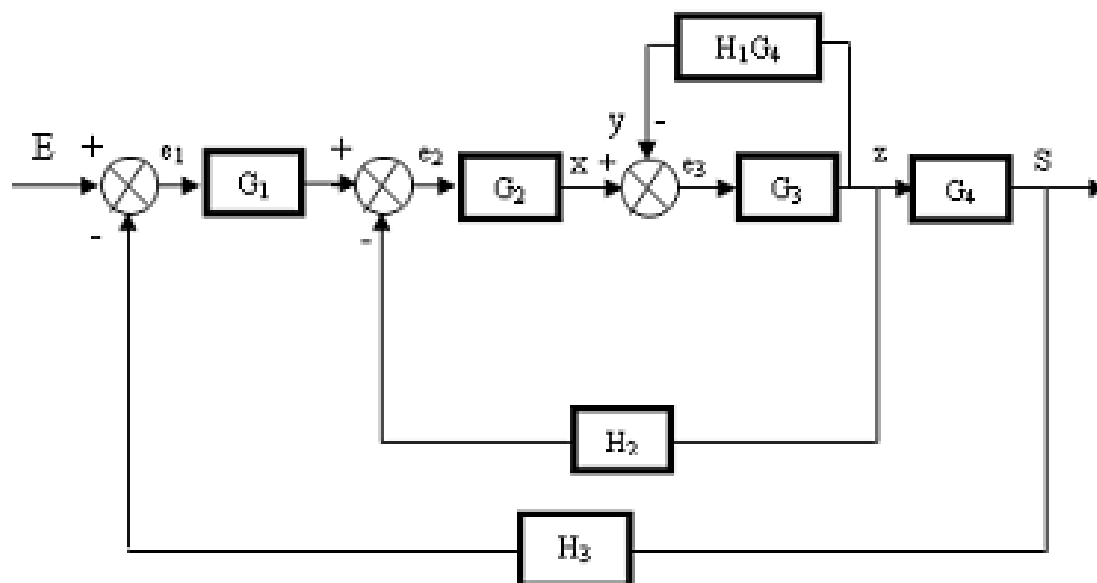
Q2 :



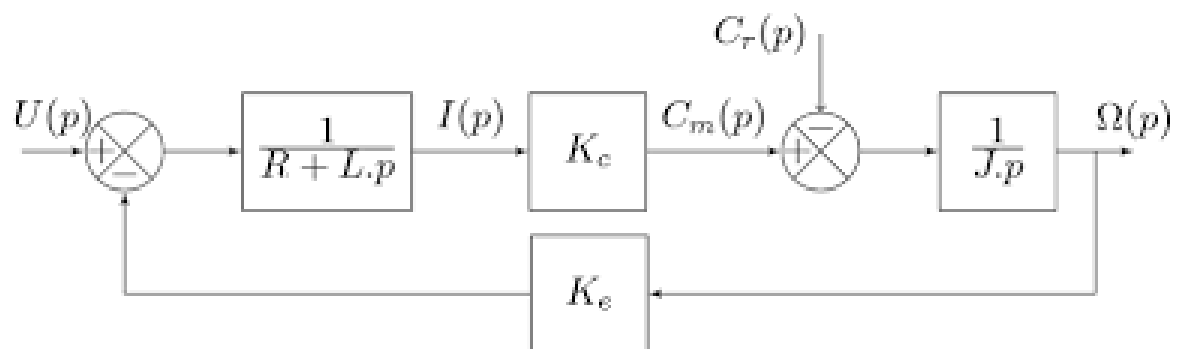
Q3 :



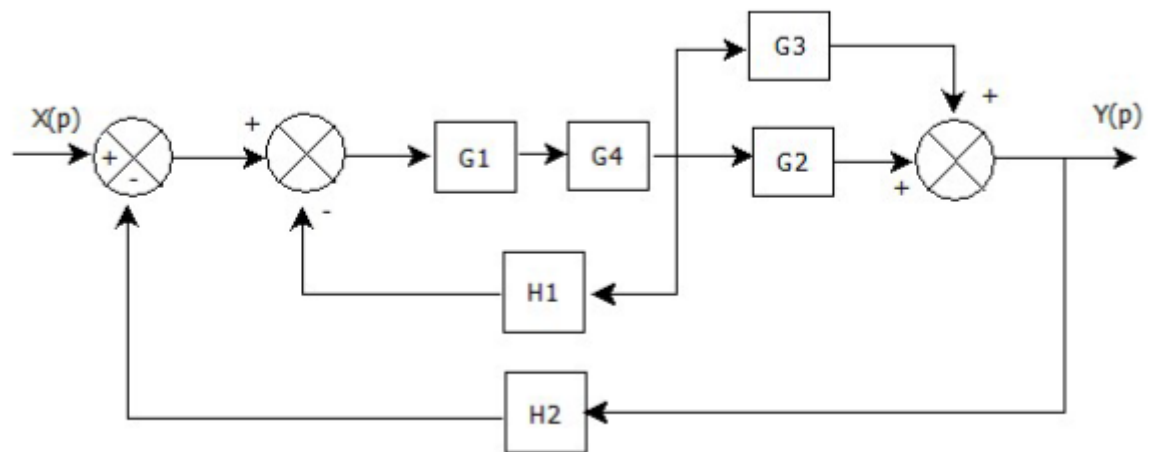
Q4 :



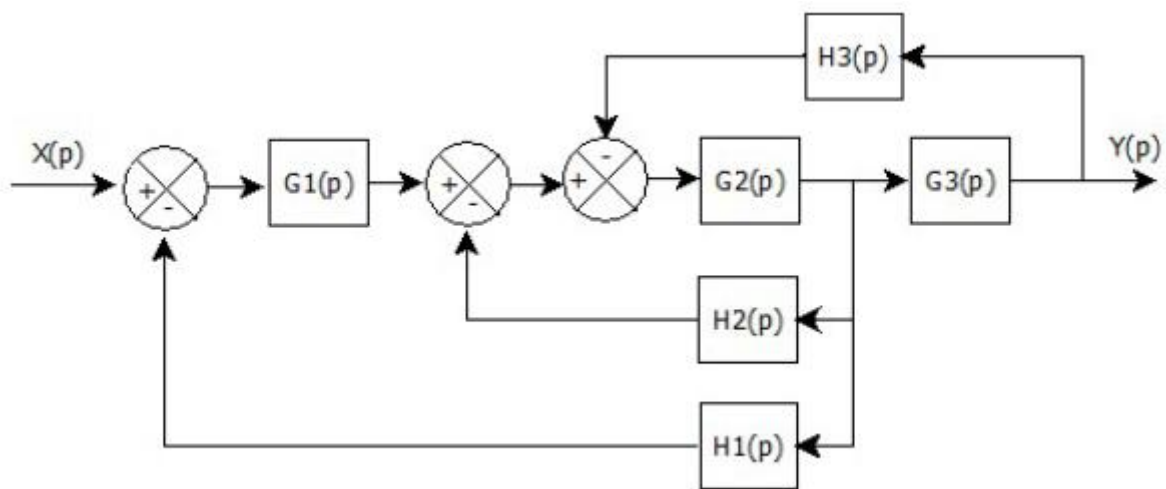
Q5 :



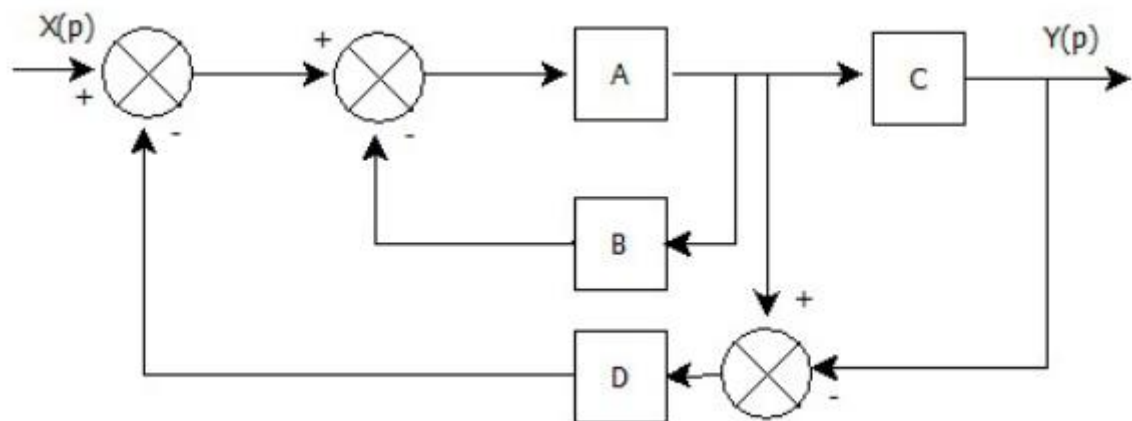
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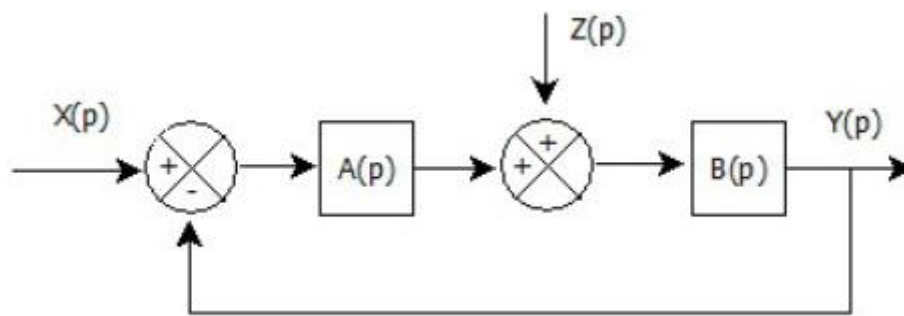
Q7 :



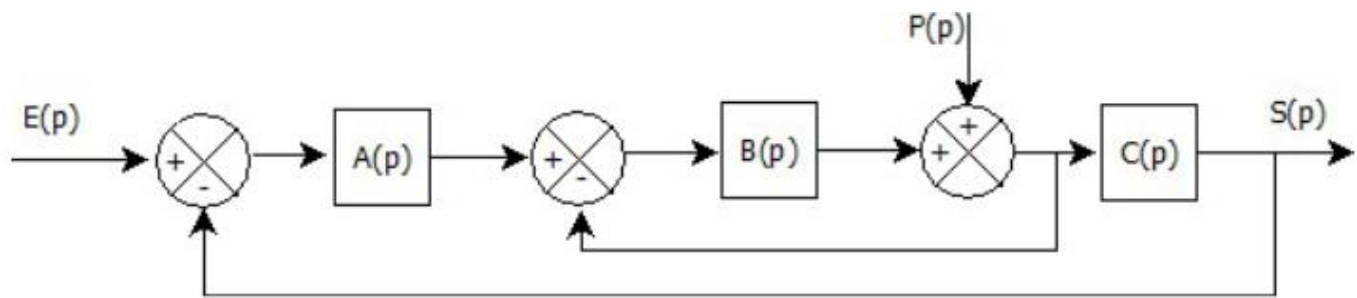
Q8 :



Q9 :



Q10 :



Q11 :

